

Session 1. What AI really is, and how an LLM writes: fluent first, right second

Friday 2 October 2026, 10:30–12:30 EDT, 120 minutes. Presentations V1 to V12 (Modules 1 and 2).

Student edition: the questions are here; the answers are discussed in the session. The Your-turn panels (sliders, tick boxes and dropdowns) live in the professor's HTML edition and are described here in words.

Questions. Because of the interpretation delay, please send questions in the Zoom chat rather than by voice, or better, afterwards on the WhatsApp number you already have. Every question, instruction and answer in this session is also written on the page, so nothing depends on hearing it in time.

How the course works

Who is in the room

First-year students of five business programmes: AI and Entrepreneurship, Fintech, Digital Marketing, Strategic Communications, Business Intelligence. No coding, no statistics, and none is needed: the only technical objects you will touch are a chat window, a token counter and a spreadsheet log.

What the course is

Twelve modules in six live sessions, two modules per session. Modules 1 to 2 today (what AI is, how an LLM writes); 3 to 4 on Wednesday (training data and bias; building a prompt one element at a time); 5 to 6 next Friday (when not to use AI; catching hallucinations); 7 to 8 (human judgement and workflows); 9 to 10 (prompt chains and quality criteria); 11 to 12 (AI policy and the capstone).

How a session runs

Flipped classroom: the university's videos and readings are watched before the session, and the session is for practice: a walkthrough of the anchor decks, a live prompt lab, a case with a poll and a debate, and a wrap-up. Today is the exception, because it is the first day: there was nothing to watch beforehand, so today's lessons teach and the practice starts after.

How the grade is built

20% attendance and participation, scored per session as full, half or no credit (present, prepared and contributing; present; absent). 20% platform work: five short quizzes after Modules 1 to 5, and the portfolio of twelve activities, five of which are graded as Assignments 1 to 5. 10% peer learning: the concept-map pair work, the live peer review of the workflow maps, the feedback forms on the capstone drafts. 50% the capstone: a prompt sequence of three or more steps with an error-recovery step, due Wednesday 4 November 2026.

The tool, the workload, and where questions go

The one tool

Open a free account on one of ChatGPT, Claude, Gemini or Copilot before Wednesday and keep the same tool for the whole course, so that your outputs can be compared from week to week. Paste every prompt and every output into your log the moment you get them: free tiers lose conversations, and the portfolio needs the exact text, not a summary.

The workload

This block is full-time: about 34 hours of work a week for three weeks. It is heavy by design. The syllabus rule is that a student who falls behind says so before the next session, not after it.

Questions

Because of the interpretation delay, questions go in the Zoom chat, or afterwards on the WhatsApp number you already have. Everything said today is also written on this page and on the slides.

Lesson 1

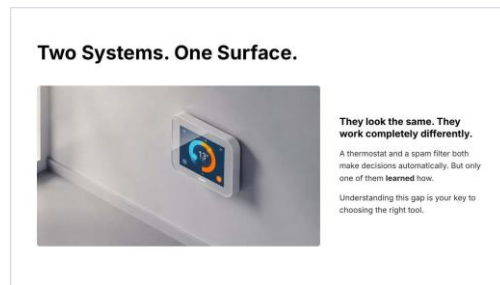
Rules or examples: the one test that tells automation from AI (artificial intelligence)

6 minutes · decks V1, V4 · shown live: V1

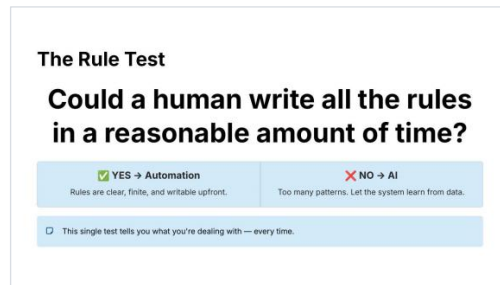
The one idea: If an engineer can write down all the rules in an afternoon, it is automation; if the system had to learn them from examples, it is AI.



V1 · page 1 · Rules vs. Learning



V1 · page 2 · Two systems, one surface



V1 · page 6 · The rule test

The one idea

If an engineer can write down all the rules in an afternoon, it is automation; if the system had to learn them from examples, it is AI.

The thermostat

Set to 20 °C. Below 20, heat on; above 20, heat off. One rule, written by a person, followed exactly, every time, including during a freak storm. Nothing is learned and nothing adapts.

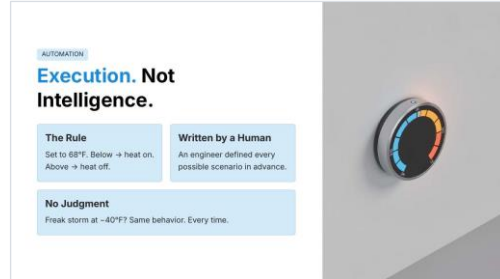
the word: automation

The spam filter

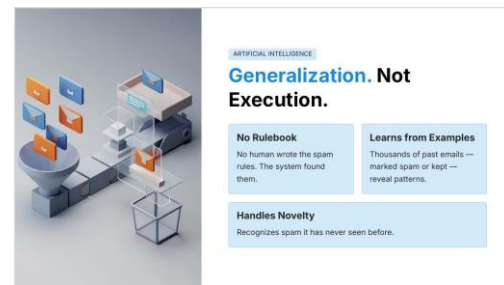
Nobody wrote a rule for every spam message. The filter studied thousands of messages that users had marked as spam or kept, found the patterns, and now sorts messages it has never seen. That ability to handle the new is what makes it AI.

the word: generalisation

Both look the same from the outside: a box that decides. The rule test looks inside: who wrote the rules, a person or the data?



V1 · page 3 · Automation: execution, not intelligence



V1 · page 4 · AI: generalisation, not execution

The Rule Test

Could a human write all the rules in a reasonable amount of time?

☒ YES → Automation Rules are clear, finite, and writable upfront.	☒ NO → AI Too many patterns. Let the system learn from data.
☐ This single test tells you what you're dealing with — every time.	

V1 · page 6 · The rule test

Story: the summer the rules stopped working (2002)

1 Rules, one at a time

In 2002 a programmer named Paul Graham spent about six months writing rules to catch spam: a word here, a sender there, a scoring point for each suspicious feature. Every rule was written by hand.

2 The rules leak

Each new rule caught a little more spam and blocked a few more real messages. Stricter rules meant more real mail lost, and losing one real message hurt more than receiving ten spam messages. The list grew and the problem did not shrink.

3 Examples instead

He changed the approach: no rules. He gave the filter two piles of his own mail, about 4,000 spam messages and about 4,000 real ones, and let it count which words appeared in which pile.

4 The result

The filter built from examples missed fewer than 5 spam messages in 1,000 and blocked none of his real mail. He published the method in August 2002 as “A Plan for Spam”, and filters built the same way spread within a year.

5 Why it kept working

When spammers changed a word to slip past the rules, the learned filter simply counted the new spelling as evidence of spam. Rules must be rewritten by a person; a filter that learns is updated by the next examples.

Source: Paul Graham, “A Plan for Spam”, August 2002 (the essay itself, checked on 30 September 2026); retold in our own words.

Three things to keep

1 Automation follows rules a person wrote; AI generalises from examples to cases it never saw.

2 The rule test: could an engineer write all the rules in a reasonable time? Yes means automation, no means AI.

3 Choosing wrong costs in both directions: rules break when the world changes; AI is wasted on a one-line rule.

Your Takeaway

Execution follows rules. Generalization learns patterns.

Apply the Rule Test to any system you build or buy.

Match the tool to the problem — not the hype.



V1 · page 15 · Your takeaway

Side by Side

Automation

- Rules written by humans upfront
- Executes instructions exactly
- Predictable, no flexibility
- Falls outside defined rules

AI

- Learns patterns from examples
- Generalizes to new situations
- Adapts to novelty and change
- Requires quality training data

V1 · page 5 · Side by side

Chat question 1 of 3

A logistics company wants a system that rings a bell when a refrigerated truck goes above 4 °C. A vendor proposes an “AI monitoring platform”. What does the rule test say?

A. AI: temperature is data, and data means machine learning.

B. Automation: the rule is one sentence and never changes, so a thermometer and an alarm are the right tool.

C. AI: the rule may need to change one day.

D. Neither: only a person can watch a truck.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

When Simple Is Better

The Boiling Water Alarm

Water reaches 212°F → bell rings.

Rule is clear, unchanging, and writable in one line.

A thermometer and alarm is all you need. No machine learning required.

The Principle

Automation is not inferior to AI. It's the **right tool** for the right job.

- Rules are clear and stable → Automate
- Patterns are complex and evolving → Use AI

Match the tool to the problem. Every time.

V1 · page 13 · When simple is better

The Rule Test

Could a human write all the rules in a reasonable amount of time?

✓ YES → Automation

Rules are clear, finite, and writable upfront.

✗ NO → AI

Too many patterns. Let the system learn from data.

☐ This single test tells you what you're dealing with — every time.

V1 · page 6 · The rule test

Chat question 2 of 3

The company inbox receives 1,000 messages a day, 300 of them spam. The rule list catches 210 spam messages this month and 150 next month, without anyone touching it. What happened?

- A. The filter got worse because rules wear out with use.
- B. The spammers changed their wording; the rules did not move, so the same rules match less.
- C. Employees marked fewer messages as spam.
- D. The rules were deleted by an update.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

The Cost of Getting It Wrong

✗ **Automation When You Needed AI**
System becomes **brittle**. Fails on anything outside the rules you wrote. Real-world complexity breaks it.

✗ **AI When Automation Would Work**
Wastes time, money, and creates **unnecessary complexity**. A simple thermometer would have done the job.

V1 · page 11 · The cost of getting it wrong

The Scheduling System That Failed



V1 · page 12 · The scheduling system that failed

Chat question 3 of 3

A vendor says its scheduling app is “AI-powered”. Inside, it blocks time slots from settings you typed: full-time staff get 40 hours, part-time get 20. Real schedules depend on availability, skills and workload, which change every week. Which sentence is right?

A. It is AI, because it decides schedules without a person.

B. It is automation, and it is the right tool, because schedules are simple.

C. It is automation sold as AI, and it will fail because the rules cannot keep up with how schedules really change.

D. It is AI, because the settings can be edited.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

The Mismatch Problem

A scheduling app claims “AI-powered” — but it’s just blocking time slots based on rules you entered.

⚠️ No learning. No adaptation. Marketing claimed AI; reality is much simpler.



V4 · page 2 · The mismatch problem

WATCH OUT

The “AI” Trap

Not AI

- Scheduling app blocking time slots by rules
- Email filter blocking obvious keywords

That’s **pattern matching**, not machine learning.



V4 · page 12 · The “AI” trap

The Scheduling System That Failed

A Real-World Cautionary Tale

A company automated scheduling with rigid rules. But real scheduling depends on **availability, skills, and shifting workloads**.

The rules couldn’t keep up. The system failed repeatedly. They needed AI.



V1 · page 12 · The scheduling system that failed

Your turn: rules or learning on the company inbox

The numbers are the inbox of the running example: 1,000 messages a day, 300 of them spam. Move the sliders to change how much the spammers change their wording, how many rules the engineer has written, and how many messages employees mark each month for the learned filter. The results panel shows how many spam messages each filter catches and how many real messages it wrongly blocks.

The controls (each with what it stands for)

- New spam wordings this month — 0 to 200, default 0. How much the spammers changed their text since the rules were written. 0 means nothing changed (the month the rules were written); 200 means almost everything did. Every 10 new wordings cost the rule list 2 points of catch rate.
- Rules written by the engineer — 20 to 400 rules, default 100 rules. The size of the rule list: forbidden words, senders and patterns. Each rule was written by hand and only matches what it was written for.
- Messages employees marked this month — 0 to 5000 marked, default 2000 marked. The examples the learned filter trains on: messages employees marked as spam or moved back to the inbox. More marks, better patterns.

Try

1. Move the new wordings from 0 to 100: this is the second month of the running example. The rule list falls from 210 to 150 caught; the learned filter stays at 285 because employees keep marking messages.
2. Put the new wordings back to 0 and raise the rules from 100 to 400: the rules now catch 285, as many as the learned filter, but they wrongly block 98 real messages instead of 35. Stricter rules lose real mail.
3. Set the marked messages to 0 and the new wordings to 200: with nothing new to learn from, the learned filter also drifts, down to 210. Learning needs examples as much as rules need an engineer.

The point: Rules are cheap and exact while the world stands still. The moment the world moves, rules need a person to rewrite them, and a filter that learns needs only the next examples.

Three things from this block

1

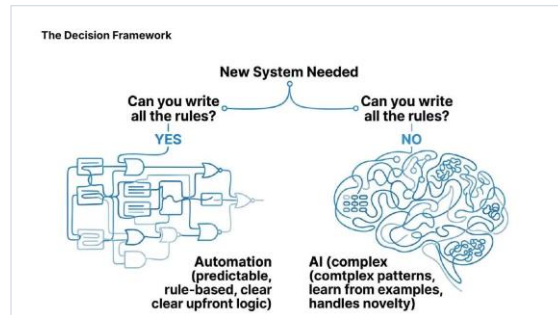
Same surface, different inside: a thermostat and a spam filter both decide, but only one learned how.

2

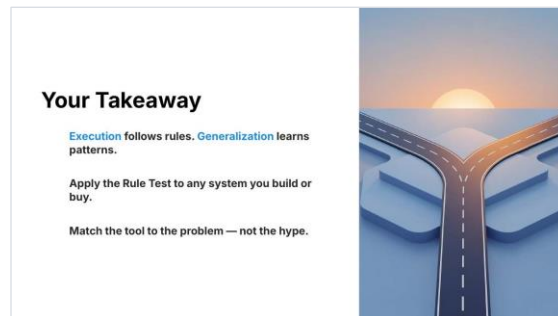
Run the rule test on every “AI” claim; the answer tells you what you are buying and how it will fail.

3

Learning needs examples: no marked messages, no improvement. Rules need a person: no engineer, no update.



V1 · page 14 · The decision framework



V1 · page 15 · Your takeaway

Lesson 2

How machines learn: the training loop, and the two ways the data breaks it

5 minutes · decks V2

The one idea: A learning system is not given the rules; it is given labelled examples, and it keeps the pattern it finds in them as a model.

How Machines Learn

Your phone predicts your next word without a single rule ever being written. That's machine learning — and it's reshaping every tool you use.



V2 · page 1 · How machines learn

The Autocomplete Mystery

No programmer wrote rules for every sentence you might type. So how does your phone know what comes next?

Not a rulebook

No if-then code for every word

Learned patterns

Trained on billions of real texts

Predicts from context

Uses what came before to guess what's next



V2 · page 2 · The autocomplete mystery

How a Machine Learning System Works



Data In

Find Patterns

Train Model

Predict

This cycle — data in, patterns found, model built, predictions made — is the heartbeat of every AI tool you use.

V2 · page 6 · How a machine learning system works

The one idea

A learning system is not given the rules; it is given labelled examples, and it keeps the pattern it finds in them as a model.

Ten thousand photos

Show a system 10,000 photos of dogs of every size, colour and pose, each labelled “dog”, and it learns what makes a dog a dog. It then recognises a puppy or a greyhound it has never seen: 94 new photos in 100 right, in the running example.

the word: generalisation

Fifty photos

Show it 50 photos and it memorises them instead. It scores 100 out of 100 on those 50 and 62 in 100 on new photos. Thin data breaks generalisation; skewed data teaches false patterns: train on faces that are 95% light-skinned and the system fails on the rest.

the word: training data

The model is the learned pattern, a long list of numbers; the software only runs it. Better data means a better model, plugged into the same software.

Learning From Examples

Feed a machine thousands of labeled emails — spam vs. real — and it finds the patterns that separate them. No rules required.



V2 · page 4 · Learning from examples

Why Volume Matters



10,000 dog photos

System learns what “dog-ness” really is — spots pugs, greyhounds, puppies it never saw before.



50 dog photos

System memorizes examples instead of learning patterns. Show it a new dog — it freezes.

V2 · page 12 · Why volume matters

When Data Is Skewed

Train on faces where 95% show light skin tones. The system learns to recognize faces — but mostly light ones. Show it a darker-skinned face, and it fails.

⚠ Bias in the data becomes bias in the model. Full stop.



V2 · page 14 · When data is skewed

Story: the checkers program that learned by playing itself (1952 to 1962)

1 A program that plays

In 1952 Arthur Samuel, an engineer at IBM, wrote a checkers program for one of the first commercial computers. The first version could play a legal game, and that was all: it did not improve.

2 It plays itself

He then let the program play thousands of games against a copy of itself. After each game it adjusted the weight it gave to features of a position (pieces ahead, control of the centre) according to whether that position had led to a win or a loss.

3 On television

On 24 February 1956 the program played on live American television. The company's president had predicted the demonstration would lift IBM's share price; it did.

4 A name for the idea

In 1959 Samuel published the results and used two words for what the program did: machine learning, the ability of a computer to improve without being explicitly programmed for every case.

5 A win, and a limit

In 1962 the program beat Robert Nealey, a strong Connecticut player, and the press called it a triumph. Three years later the world champion, Walter Hellman, won all four games he played against it by post and drew a fifth in person. More examples had made it good; they had not made it the best.

Sources: IBM, "The games that helped AI evolve"; J. R. Delaney, "The man who taught machines to learn" (both checked 30 September 2026); retold in our own words.

Three things to keep

1

The loop: labelled data in, patterns found, a model built, predictions made, and repeat with more data.

2

The model is the learned pattern, not the software; a better model is trained and plugged in, not coded.

3

Thin data makes the system memorise; skewed data teaches it a false pattern; only balanced, plentiful, clean data generalises.

The Key Takeaways

01

Rules vs. patterns

Traditional programming writes explicit rules. Machine learning finds hidden ones from labeled examples.

03

Data quality is everything

Thin, skewed, or noisy data isn't a software bug — it's a fundamental flaw in what the model learned.

02

The training loop

Data → patterns → model → predictions → repeat. Every AI tool runs on this same cycle.

04

Generalization is the goal

A good model handles new, unseen situations — not just the examples it was trained on.

V2 · page 23 · The key takeaways

Machine Learning in One Sentence

Find patterns in labeled data. Apply them to the unknown.

That's the entire idea — and it's running inside every AI tool you use today.

V2 · page 25 · Machine learning in one sentence

Chat question 1 of 3

A retailer trained a photo sorter on 50 labelled product photos. It scores 100 in 100 on those photos and 62 in 100 on new ones. What is happening?

- A. The software has a bug that appears on new photos.
- B. The system memorised the 50 examples instead of learning the pattern; the data is too thin to generalise.
- C. New photos are harder than old photos.
- D. 62 in 100 is the best any system can do on photos.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

Thin Data Breaks Generalization

Too few examples

A system trained on too little data memorizes, it doesn't learn. It fails the moment it sees anything new.

V2 · page 13 · Thin data breaks generalisation

Why Volume Matters



10,000 dog photos

System learns what "dog-ness" really is — spots pugs, greyhounds, puppies it never saw before.



50 dog photos

System memorizes examples instead of learning patterns. Show it a new dog — it freezes.

V2 · page 12 · Why volume matters

Chat question 2 of 3

A face-detection tool was trained on photos of which 95% show light skin tones. In use it works on some faces and fails on others. Which sentence is right?

A. It is a software bug that a patch will fix.

B. The training data was skewed, so the model learned the group it saw most; the fix is balanced data and retraining.

C. The tool needs more photos of the same kind.

D. Faces cannot be detected by machine learning.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

When Data Is Skewed

Train on faces where 95% show light skin tones. The system learns to recognize faces — but mostly light ones. Show it a darker-skinned face, and it fails.

⚠ Bias in the data becomes bias in the model. Full stop.



V2 · page 14 · When data is skewed

Real Data, Real Consequences

What happened

A major tech company deployed a face-detection tool that performed well on some faces and poorly on others.

Root cause: Imbalanced training data.

The lesson

This wasn't a software bug. It was a data problem — and no code patch could fix it.

Data quality is a design decision, not an afterthought.

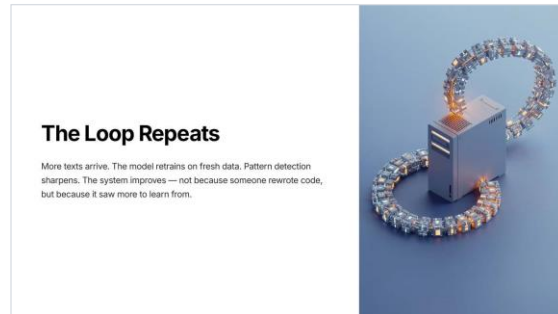
V2 · page 15 · Real data, real consequences

Chat question 3 of 3

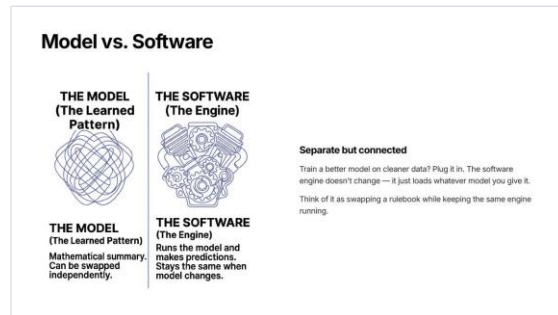
The marketing team adds 3,000 new labelled photos to the shared folder on Monday. On Tuesday the photo sorter makes the same mistakes as before. Why?

- A. The new photos are of poor quality.
- B. The model has not been retrained; the software runs the old model until a new one is trained and plugged in.
- C. The software needs to be rewritten to use more photos.
- D. 3,000 photos are too few to matter.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.



V2 · page 11 · The loop repeats



V2 · page 17 · Model versus software

Your turn: feed the loop, then retrain

The numbers are the photo sorter of the running example. Choose how many labelled photos the system trains on and what share of them come from group A, then press Retrain to run the loop. The results panel shows the score on the training photos, on new photos, and on new photos of each group. Until you press Retrain, the sliders change nothing: the software keeps running the old model.

The controls (each with what it stands for)

- Labelled photos available for training — options: 50 photos; 100 photos; 200 photos; 500 photos; 1,000 photos; 2,000 photos; 5,000 photos; 10,000 photos. How many examples the loop will learn from when you retrain. Each photo carries its label (what is in it).
- Share of the photos that come from group A — 50 to 95%, default 50%. 50% means the two groups are equally present; 95% means the system almost only sees group A. The rest come from group B.
- Retrain the model (run the loop, plug in the new model) — a button. Runs the training loop on the photos and share chosen above and replaces the model the software is running. Nothing else changes the numbers.

Try

1. Choose 50 photos and press Retrain: 100 on the training photos, 62 on new ones. The gap of 38 points is the sign of memorising.
2. Choose 10,000 photos, leave the share at 50%, and press Retrain: 94 on new photos, the same for both groups.
3. Keep 10,000 photos, move the share to 95%, and read the panel before pressing Retrain: nothing moved. Press Retrain: group A rises to 97 and group B falls to 70.

The point: Volume raises the score, skew moves it between groups, and none of it happens until the loop runs again. The figures are fixed teaching values chosen to behave like real runs.

Three things from this block

1 Traditional software is rules written by a developer; machine learning is patterns found in labelled examples.

2 Every learning system runs the same loop, and it improves from more data, not from someone rewriting the code.

3 Thin data memorises, skewed data misleads; the fixes are more data, balanced data, cleaner data, and retraining.

The Key Takeaways

01

Rules vs. patterns

Traditional programming writes explicit rules. Machine learning finds hidden ones from labeled examples.

03

Data quality is everything

Thin, skewed, or noisy data isn't a software bug — it's a fundamental flaw in what the model learned.

02

The training loop

Data → patterns → model → predictions → repeat. Every AI tool runs on this same cycle.

04

Generalization is the goal

A good model handles new, unseen situations — not just the examples it was trained on.

V2 · page 23 · The key takeaways

Three Data Failure Modes



Thin Data

Too few examples → system memorizes instead of generalizes



Skewed Data

Unbalanced examples → system learns false or partial patterns



Noisy Data

Errors in labels → model learns from mistakes and misfires on real inputs

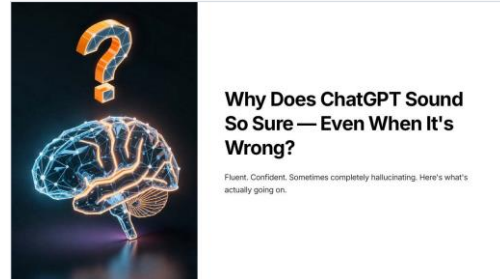
V2 · page 16 · Three data failure modes

Lesson 3

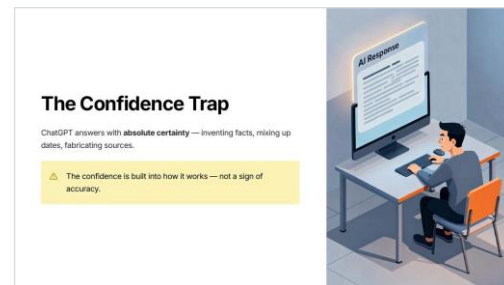
The family tree: an LLM (large language model) is one branch of machine learning, and often not the right one

6 minutes · decks V3 · shown live: V3

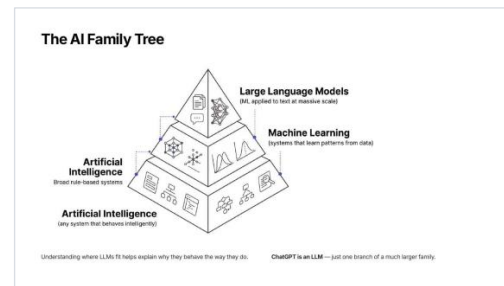
The one idea: AI is the widest circle, machine learning sits inside it, and a large language model is one branch inside that: a text predictor made large.



V3 · page 1 · Why does ChatGPT sound so sure, even when it's wrong?



V3 · page 2 · The confidence trap



V3 · page 3 · The AI family tree

The one idea

AI is the widest circle, machine learning sits inside it, and a large language model is one branch inside that: a text predictor made large.

One job, made large

An LLM does one thing: given some text, it predicts the next token, then the next. What makes it “large” is scale: GPT-3 has 175 billion adjustable numbers, against about a million in an older autocomplete and thousands in a spell-checker, and it was trained on billions of tokens of text for weeks on thousands of chips.

the word: parameters

The siblings

Image classifiers recognise objects in photos, recommendation engines rank what to show you, fraud detectors flag a payment. Each is machine learning trained on different data for a different task, and each is usually cheaper, faster and more exact at its own job than a language model would be.

the word: machine learning

The LLM gets the headlines because it is general-purpose and answers in readable text. It is a tool with specific strengths, inside a family with other tools.

What Is Artificial Intelligence?



The Broadest Category

Any system that **behaves intelligently**—whether following hand-coded rules or learning from data.

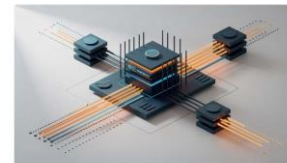
The key is the behavior, not how it's built.

V3 · page 4 · What is artificial intelligence?

Machine Learning: Learning from Data

No manual rules

ML systems get better by studying **examples**, not by being explicitly programmed for every scenario.



V3 · page 5 · Machine learning: learning from data

What Makes an LLM an LLM?

Core Task

Predicts the **next word** in a sequence — one word at a time

The “Large” Part

GPT-3: **175 billion parameters** vs. millions in autocomplete

The Scale Secret

Trained on **billions of text tokens** over weeks on thousands of chips



V3 · page 6 · What makes an LLM an LLM?

Story: how the family got its names (1955 to 2022)

1 1955: a word is chosen

In a proposal for a summer workshop at Dartmouth College, John McCarthy and three colleagues (Marvin Minsky, Nathaniel Rochester and Claude Shannon) wrote the phrase “artificial intelligence” for the first time. Their bet: every feature of intelligence can be described precisely enough for a machine to simulate it.

2 1959: the inner circle

Arthur Samuel, whose checkers program learned by playing itself, named the subset that learns from examples rather than from written rules: machine learning. AI was the behaviour; machine learning was one way to get it.

3 2017: a new design for text

A team at Google published a design called the transformer, which lets a model weigh every word of a text against every other at once. It became the design behind the language models that followed.

4 2020: large

OpenAI’s GPT-3 was trained with 175 billion adjustable numbers on a large slice of the internet’s text. The word “large” in “large language model” dates from this jump in scale.

5 2022: the branch everyone sees

On 30 November 2022 ChatGPT put a language model behind a chat window. It reached an estimated 100 million users within two months, and for most people the newest branch became the whole tree.

Sources: Wikipedia, “Dartmouth workshop” and “Attention Is All You Need”; the GPT-3 paper (“Language Models are Few-Shot Learners”, 2020); AI Business on the UBS user estimate (checked 30 September 2026); retold in our own words.

Three things to keep

1

AI is any system that behaves intelligently; ML is the part that learns from data; an LLM is ML applied to text at a very large scale.

2

An LLM predicts the next token; “large” means billions of parameters and billions of tokens of training text.

3

The siblings (image classifiers, recommenders, fraud detectors) are often cheaper, faster and more exact for their own job.

Key Takeaways

- 1 LLMs are one branch of AI**
Nested inside Machine Learning,
inside Artificial Intelligence
- 2 Scale is the secret**
Billions of parameters + billions
of tokens = emergent capability
- 3 Confidence ≠ correctness**
No truth-checking — pattern
prediction all the way down

V3 · page 14 · Key takeaways

175 Billion Parameters

175B

GPT-3 Parameters

Adjustable numbers used to make
predictions

~1M

Autocomplete

Parameters in earlier text tools

The scale is the secret.

1000s

Chips Required

Specialized chips running for weeks to
train one model

V3 · page 7 · 175 billion parameters

Chat question 1 of 3

A bank must decide in 0.1 second whether to approve a card payment, 1,000,000 times a day. The innovation team proposes to ask a language model each time. Using the price list of the running example, what is the best answer?

- A. Use the LLM: it is the most advanced model available.
- B. Use a fraud classifier: it answers in 0.05 s for \$100 a day, while the LLM would take 2 s and cost \$10,000 a day.
- C. Use the LLM but only during office hours.
- D. Neither; card payments cannot be checked by machine learning.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

LLMs Have Siblings



V3 · page 8 · LLMs have siblings

Right Tool for the Right Job



LLMs aren't always the answer

A streaming recommendation engine needs to be **quick and efficient** — an LLM would be overkill.

Understanding the whole ML family helps you **pick the right tool**.

V3 · page 13 · Right tool for the right job

Chat question 2 of 3

A colleague says: “ChatGPT is AI, so it must be able to recognise which of our product photos are damaged.” Which reply uses the family tree correctly?

A. Yes: any AI can do any AI task.

B. A language model predicts text; recognising damage in photos is the job of an image classifier, a different branch of machine learning.

C. No: ChatGPT is not AI.

D. Yes, if the photos are described in words first.

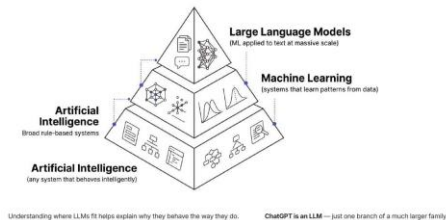
Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

LLMs Have Siblings



V3 · page 8 · LLMs have siblings

The AI Family Tree



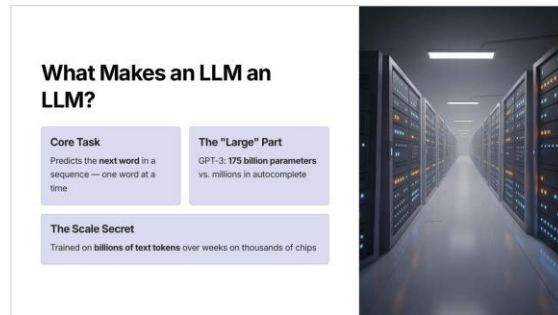
V3 · page 3 · The AI family tree

Chat question 3 of 3

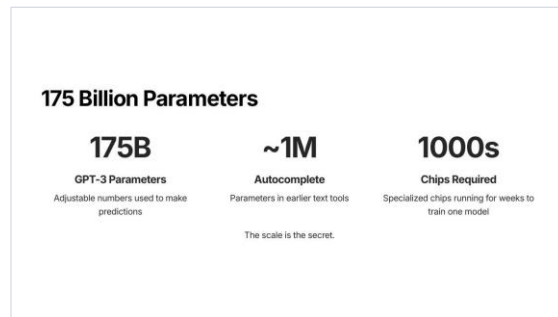
Why does the deck say “scale is the secret” of a large language model?

- A. Because it has a database of facts that is very large.
- B. Because it reasons like a person, only faster.
- C. Because it predicts the next token with 175 billion adjustable numbers trained on billions of tokens of text, and that scale is what makes the fluent text possible.
- D. Because it is updated every day with new text.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.



V3 · page 6 · What makes an LLM an LLM?



V3 · page 7 · 175 billion parameters

Your turn: pick the family member, then count the cost

The numbers are the price list of the running example: \$0.01 and 2 seconds per LLM request; \$0.0001 and 0.05 seconds per request to a small specialised model. Choose a task, how many requests a day it gets, and how long each request may take. The results panel names the family member that fits and shows the daily cost and the delay for that tool and for an LLM used anyway.

The controls (each with what it stands for)

- The task — options: Write a reply to a customer complaint; Choose which product to show first on the home page; Flag a suspicious card payment; Find the faces in 10,000 photos; Summarise a 20-page report in one page; Predict next month's sales from five years of monthly figures. What the system must produce: text (a reply, a summary), a ranking, a category (fraud or not; face or not) or a number (a forecast). The output decides the branch.
- Requests per day — options: 100; 1,000; 10,000; 100,000; 1,000,000. How many times a day the system is asked. Cost per day is requests multiplied by the price per request.
- Longest acceptable delay per request — options: 0.1 second (a card terminal); 1 second (a web page); 10 seconds (an e-mail draft). How long the person or the system at the other end can wait. An LLM takes about 2 seconds per request in this example; a small model about 0.05.

Try

1. Choose “Flag a suspicious card payment”, 1,000,000 requests and 0.1 second: the fitting tool costs \$100 a day; an LLM would cost \$10,000 and be twenty times too slow.
2. Choose “Write a reply to a customer complaint” at 10,000 requests and 10 seconds: an LLM is the fitting tool at about \$100 a day, and the delay fits.
3. Keep the reply task and set the delay to 0.1 second: the LLM is now too slow for the task as designed; the answer is to change the step (draft in the background, show it later), not to switch branch.

The point: The output of the task chooses the branch; the volume and the allowed delay tell you what the choice costs. The price list is a fixed teaching value chosen to behave like real systems.

Three things from this block

1

Three nested circles: AI, then machine learning, then the language model; “ChatGPT is AI” is true and says almost nothing.

2

“Large” is a statement about scale, not about a new kind of thinking: billions of parameters, billions of tokens.

3

When the output is not text, a sibling is usually the cheaper, faster and more exact tool; ask what the task produces.

Key Takeaways

- 1 LLMs are one branch of AI
Nested inside Machine Learning,
inside Artificial Intelligence
- 2 Scale is the secret
Billions of parameters = billions
of tokens = emergent capability
- 3 Confidence ≠ correctness
No truth-checking — pattern
prediction all the way down

V3 · page 14 · Key takeaways

No truth-checking system exists.

An LLM predicts what words usually come next — it has no separate mechanism to verify whether something is **actually true**.

The confidence is a feature of the design, not a signal of accuracy.

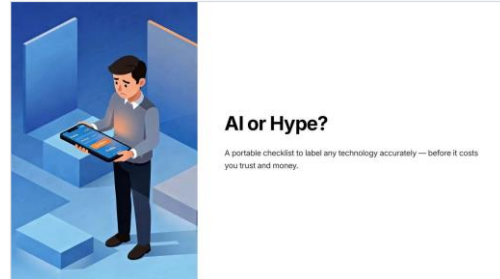
V3 · page 12 · No truth-checking system exists

Lesson 4

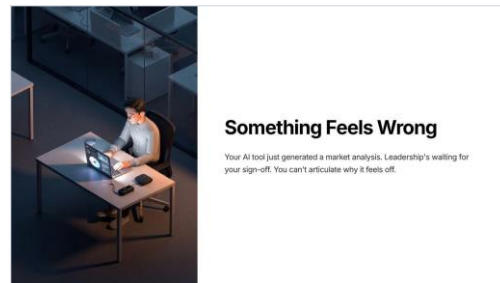
AI or hype: three questions for any claim, three misconceptions to drop, and three words to replace

5 minutes · decks V4, V5, V6

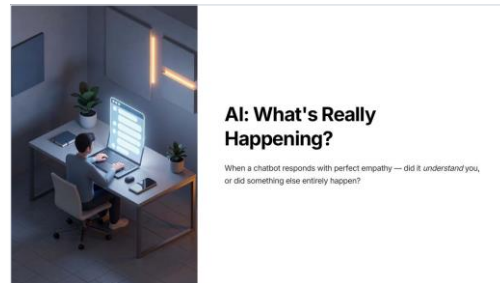
The one idea: Ask what data it trains on, what task it generalises to and whether it updates; if the answers are missing, the label is marketing, and if the answers are there, say “the model predicts”, never “the AI thinks”.



V4 · page 1 · AI or hype?



V5 · page 1 · Something feels wrong



V6 · page 1 · AI: what's really happening?

The one idea

Ask what data it trains on, what task it generalises to and whether it updates; if the answers are missing, the label is marketing, and if the answers are there, say “the model predicts”, never “the AI thinks”.

The scheduling app

Sold as “AI-powered”. Inside: it blocks time slots from settings you typed. No training data, no generalisation, no updating. Three questions, three empty answers: automation with a fashionable label. Vagueness is the tell; a testable claim names the data, the task and the technique.

the word: automation

The empathetic chatbot

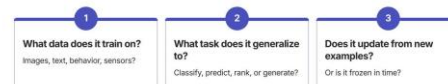
It answers a complaint with perfectly timed sympathy. It did not understand or feel anything: it learned, from millions of conversations, which phrases follow a complaint. We complete the illusion ourselves, as we do with puppets and dolls. Narrow, not general; pattern completion, not sentence.

the word: narrow AI

The word swap protects your judgement: “the model predicts” instead of “the AI thinks”, “has learned a pattern for” instead of “understands”. If the sentence stops making sense about a formula, you had slipped.

Your Portable Toolkit

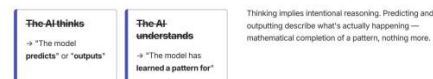
Three questions. Any product. Any claim.



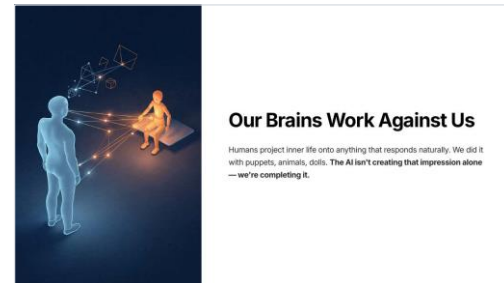
V4 · page 4 · Your portable toolkit: three questions

FIX YOUR LANGUAGE

Replace These Words — Right Now



V6 · page 13 · Replace these words, right now



V6 · page 9 · Our brains work against us

Story: the program that only rephrased your words, and the secretary who asked to be left alone with it (1966)

1 A very short rulebook

In the mid-1960s Joseph Weizenbaum, a professor at MIT, wrote a small program called ELIZA. It found a keyword in what you typed, applied a rule, and turned your sentence into a question. “I am unhappy” became “Why are you unhappy?” It knew nothing about anything.

2 Playing the therapist

Its best-known script imitated a therapist who reflects your words back to you. That style needs no knowledge of the world, so the program could keep a conversation going for a long time with a few dozen rules.

3 The secretary

Weizenbaum’s own secretary, who had watched him build the program and knew it was rules, asked him to leave the room so that she could talk to it in private. He wrote later that he had not realised how quickly a simple program could make normal people believe it understood them.

4 A warning in a book

He spent the next decade warning against confusing a program’s answers with understanding, and in 1976 published a book, “Computer Power and Human Reason”, about the difference between calculating and judging.

5 The ELIZA effect

The habit of reading understanding and feeling into a machine that answers in words now carries the program’s name: the ELIZA effect. Today’s chatbots are not rulebooks, but the habit is the same, and V6’s word swap is the cure.

Source: Wikipedia, “ELIZA” (the 1966 paper in Communications of the ACM, the secretary anecdote and the 1976 book), checked 30 September 2026; retold in our own words.

Three things to keep

1 Three questions unmask any claim: what data does it train on, what task does it generalise to, does it update from new examples?

2 Today's systems are narrow: excellent at one bounded task, unable to carry the skill elsewhere; general intelligence does not exist, and no system has any confirmed sentience.

3 Say “the model predicts” and “has learned a pattern for”; the formula test catches the slip (“the formula understands” makes no sense).

Red Flag Phrases

"Uses artificial intelligence"

...without naming the specific task or data source

No technique named

Legitimate AI claims name the method, describe the data, explain the task.

Vagueness is the tell

If they can't answer what it trains on — it's probably not real ML.

V4 · page 13 · Red-flag phrases

Key Takeaways

01

Narrow AI exists. General AI does not. Sentience: unconfirmed.

02

Chatbots pattern-match at scale — not empathy, not understanding.

03

Precise language protects your credibility and your deployments.

04

Audit rigorously — knowing it's pattern-matching changes how you inspect.

V6 · page 19 · Key takeaways

Chat question 1 of 3

A vendor's brochure says: "Our platform uses artificial intelligence to optimise your invoices." You ask what data it trains on and what task it performs. The salesperson answers with more adjectives. What should you conclude?

- A. The system is too advanced to explain to a buyer.
- B. The claim is not testable: a real machine-learning claim names the data, the task and the technique, and technical teams can answer in a minute.
- C. Invoices cannot be processed by AI.
- D. The vendor is hiding a competitor's technology.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

Red Flag Phrases

"Uses artificial intelligence"

...without naming the specific task or data source

No technique named

Legitimate AI claims name the method, describe the data, explain the task

Vagueness is the tell

If they can't answer what it trains on — it's probably not real ML

V4 · page 13 · Red-flag phrases

How to Test Any Claim

■ "What data do you train on?"

■ "Supervised or unsupervised?"

■ "What task does it generalize to?"

If they deflect with marketing-speak — it's probably not real machine learning.



V4 · page 14 · How to test any claim

Chat question 2 of 3

A team lead writes in a report: “The AI understood our customers’ frustration and answered with empathy.” Rewrite the sentence with V6’s vocabulary.

A. “The AI felt our customers’ frustration and chose to be kind.”

B. “The model has learned a pattern for sympathetic replies from millions of conversations and predicted one; whether it fits our customers is for us to check.”

C. “The AI is sentient and should be treated as a colleague.”

D. “The AI thinks our customers are frustrated.”

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

FIX YOUR LANGUAGE

Replace These Words — Right Now

The AI thinks → “The model predicts” or “outputs”	The AI understands → “The model has learned a pattern for”
---	--

Thinking implies intentional reasoning. Predicting and outputting describe what’s actually happening — mathematical completion of a pattern, nothing more.

V6 · page 13 · Replace these words, right now



Statistical Mimicry

A model trained on millions of empathetic conversations learns to output sympathetic phrases.

Not because it feels your pain — because that pattern completed successfully millions of times.

V6 · page 8 · Statistical mimicry

Chat question 3 of 3

A manager audits the outputs of a recommendation engine less often than before, saying “it understands our catalogue now”. Which V6 cost is this?

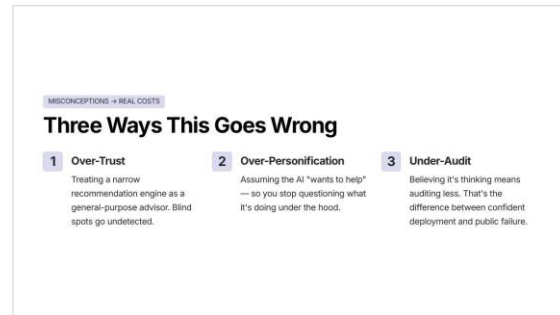
A. Over-trust: a narrow tool is treated as a general adviser.

B. Under-audit: believing the system thinks, the manager checks it less, so errors outside its training data slip through.

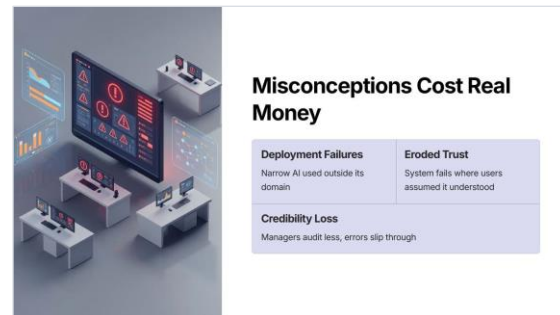
C. No cost: recommendation engines do not make errors.

D. Over-personification, which only affects chatbots.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.



V6 · page 11 · Three ways this goes wrong



V6 · page 12 · Misconceptions cost real money

Your turn: the three questions on a vendor claim

The material is the running example: six product claims. Pick a claim, then answer V4's three questions as the vendor did in the meeting (or leave "not stated" when the vendor did not answer). The results panel gives the label your answers imply, a vagueness meter (how many of the three questions are unanswered), and the answer key. The plain-words switch rewrites the claim with V6's vocabulary.

The controls (each with what it stands for)

- The claim — options: "Our AI-powered scheduler plans your team's week for you."; "A smart thermostat that understands your comfort."; "Our AI writing assistant thinks through your e-mails with you."; "AI fraud alerts that know when a payment is not you."; "AI-driven e-mail protection."; "A support chatbot that truly understands how your customers feel.". The sentence as the vendor wrote it. The words thinks, understands and knows are the ones V6 asks you to replace.
- Question 1: what data does it train on? — options: Not stated by the vendor; None: the rules were typed by a person; Your behaviour (clicks, listening, purchases); Labelled examples (payments marked fraud or not, photos with tags); Large collections of text. Training data is what the system learned from. "None" means it learned nothing: it follows rules.
- Question 2: what task does it generalise to? — options: Not stated by the vendor; Follow a fixed rule; Rank or recommend; Classify or flag; Generate text. What the system produces for a new case it never saw. A fixed rule does not generalise; it repeats.
- Question 3: does it update from new examples? — options: Not stated by the vendor; No: frozen rules; Yes, from new examples as they arrive; Between versions, when the vendor retrains. Whether new examples change the system, and when. Rules never update by themselves; models update when retrained.
- Plain-words mode: rewrite the claim with V6's vocabulary — a tick box, off by default. Replaces thinks, understands and knows by what the mechanism does: follows a rule, predicts, has learned a pattern for.

Try

1. Keep the scheduler claim and answer the three questions as the truth is (none, follow a fixed rule, no): the label is automation sold as AI, with vagueness 0.
2. Pick the fraud-alert claim and leave all three questions as "not stated": no label, vagueness 3 of 3. Then answer them (labelled examples, classify or flag, yes): machine learning, vagueness 0.
3. Pick the empathetic chatbot, tick plain-words mode, and read the rewritten sentence aloud: it says what happened and what remains to be checked.

The point: A claim becomes testable the moment the three questions are answered; a sentence becomes honest the moment the verbs describe the mechanism. The claims are written for teaching and name no company.

Three things from this block

1

Before accepting any “AI” label, ask what the system does with data; marketing lags behind reality and your checklist keeps you ahead.

2

Three misconceptions cost money: conscious, general, understanding; the tools are narrow statistical engines with no inner life.

3

Vocabulary is your shield: predicts, outputs, has learned a pattern for; reserve sentient, conscious and aware for evidence.

Key Takeaways

Not conscious

AI is a statistical pattern-matcher, not a thinker

Not general

AGI doesn't exist yet — current tools are narrow

Not understanding

Fluent output ≠ accurate output

Next up

Five core concepts every professional should know

V5 · page 15 · Key takeaways

Key Takeaways

01

Narrow AI exists. General AI does not. Sentence: unconfirmed.

02

Chatbots pattern-match at scale — not empathy, not understanding.

03

Precise language protects your credibility and your deployments.

04

Audit rigorously — knowing it's pattern-matching changes how you inspect.

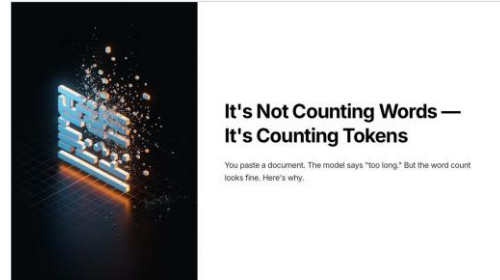
V6 · page 19 · Key takeaways

Lesson 5

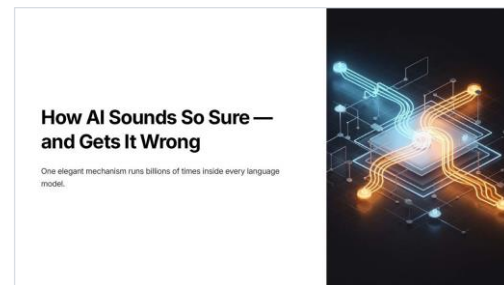
Tokens, the context window, and the one operation behind every answer: predict the next token

7 minutes · decks V7, V8 · shown live: V7, V8

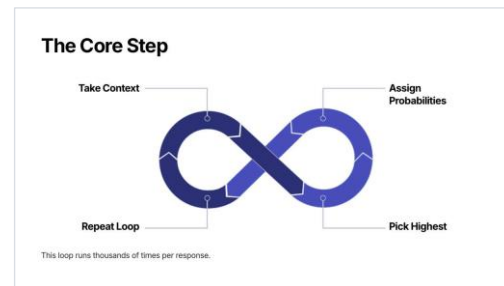
The one idea: The model reads and writes tokens, not words; it gives every possible next token a probability and writes the most likely one, thousands of times per answer, with no fact-checker anywhere in the loop.



V7 · page 1 · It's not counting words, it's counting tokens



V8 · page 1 · How AI sounds so sure, and gets it wrong



V8 · page 4 · The core step

The one idea

The model reads and writes tokens, not words; it gives every possible next token a probability and writes the most likely one, thousands of times per answer, with no fact-checker anywhere in the loop.

Counting tokens

“Unhappiness” is one word and three tokens: un, happ, iness. A space can be a token; so can a comma. About four characters or three quarters of a word make a token, so 100 words are about 133 tokens and a 10,000-word document is about 13,300: more than an 8,000-token window can hold. Plain words cost fewer tokens than rare ones.

the word: token

The capital of France is

The model has seen that phrase end in “Paris” thousands of times and in “Lyon” a few. So it gives Paris 95%, Lyon 3%, everything else 2%, writes Paris, adds it to the context, and repeats. It does not know that Paris is a city. If a false claim had been frequent in the training text, it would be written with the same confidence.

the word: next-token prediction

This is why prompt engineering works: the tokens you put in the prompt change which patterns the model treats as relevant, and so change the probabilities of what comes out.

One Word, Three Tokens

“unhappiness” → “un” + “happ” + “iness”

Tokens are smaller than words — and they’re what the model actually sees.



V7 · page 3 · One word, three tokens

The Mental Rule of Thumb

1 token ≈ 4 characters	1 token ≈ 0.75 words	100 words ≈ 130–150 tokens
---------------------------	-------------------------	-------------------------------

Close enough to guide real decisions — not exact, but useful.

V7 · page 7 · The mental rule of thumb

A Concrete Example

“The capital of France is ____”

95%

Paris

Appeared thousands of times in training data

3%

Lyon

Appeared far less frequently

2%

Everything else

Rare patterns, low probability

V8 · page 6 · A concrete example

Story: the guessing game at the Shannons' kitchen table (1951)

1 A book off the shelf

Around 1950 Claude Shannon, the engineer who founded information theory, took a book down at home, hid the title, chose a passage, and asked his wife Betty to guess it one letter at a time. Right, he said, or wrong, try again, until she had the letter.

2 Twenty-seven symbols

The alphabet of the game had 27 symbols: 26 letters and the space. For each letter she used what came before it: the words so far, the sentence, the style of the text.

3 Sixty-nine percent

In the sample he published, 89 of 129 letters were guessed right on the first try, 69%. Common continuations were easy (after “q”, a “u”; after “th”, an “e”); rare words and the first letters of new sentences were hard.

4 A number for English

In 1951 he published the experiment in “Prediction and Entropy of Printed English”. With 100 letters of context, a reader’s uncertainty about the next letter came down to about one bit: English is predictable enough that most of the next letter is already in the text before it.

5 The same game, at scale

A language model plays Betty’s game with tokens instead of letters and billions of examples instead of one reader’s experience. It scores far better than 69%, and it fails the same way she did: on rare words and on things that are not in the text so far.

Sources: C. E. Shannon, “Prediction and Entropy of Printed English”, Bell System Technical Journal, January 1951 (the paper); LessWrong, “A conceptual precursor to today’s language machines” (the kitchen-table account); both checked 30 September 2026; retold in our own words.

Three things to keep

1

Tokens, not words: about four characters or three quarters of a word each; the window holds prompt, history and answer together.

2

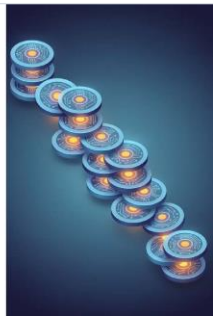
One operation, repeated: take the text so far, give every token a probability, write the most likely one, add it, repeat.

3

High probability means common in the training text, not true; a frequent falsehood is written as confidently as a fact.

The Full Picture

- 1 Next-token prediction**
Repeated thousands of times per response
- 2 No magic, no database, no reasoning**
Probability assignment + greedy selection, cascading forward



V8 · page 12 · The full picture

What You Now Know

- 1 token $\approx$ 4 chars / 0.75 words
- Fancy words cost more tokens
- Spaces and punctuation count
- Contractions are unpredictable

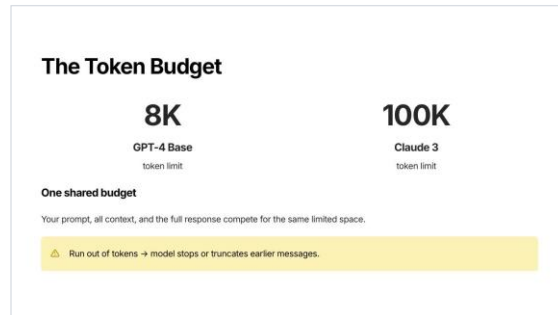
V7 · page 16 · What you now know

Chat question 1 of 3

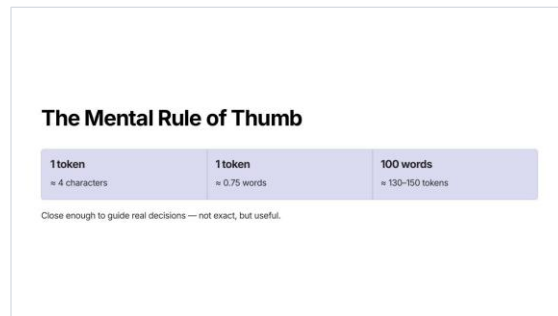
A manager pastes a 10,000-word report into a tool whose window is 8,000 tokens and asks for a summary. What happens, and what is the arithmetic?

- A. It fits: 10,000 words is less than 8,000 tokens because tokens are bigger than words.
- B. It does not fit: 10,000 words are about 13,300 tokens, so the tool refuses or drops the start, and nothing is left for the answer.
- C. It fits if the manager writes “please summarise” politely.
- D. The tool reads the first 8,000 words and summarises those, which is fine.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.



V7 · page 5 · The token budget



V7 · page 7 · The mental rule of thumb

Chat question 2 of 3

After “The capital of France is”, the model gives Paris 95% and Lyon 3% and writes Paris. What does the 95% mean?

- A. The model checked a map and is 95% sure Paris is correct.
- B. “Paris” followed that phrase far more often than any other word in the training text; the number is about frequency, not truth.
- C. There is a 5% chance that Lyon is the capital of France.
- D. The model reasons about geography and concludes Paris.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

A Concrete Example

“The capital of France is ____”

95%

Paris

Appeared thousands of times in training data

3%

Lyon

Appeared far less frequently

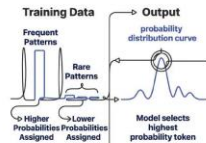
2%

Everything else

Rare patterns, low probability

V8 · page 6 · A concrete example

Pattern Frequency → Probability



The key insight

The model learned: when these tokens appear, these others tend to follow.
Frequency in training data becomes probability. That's the whole rule.

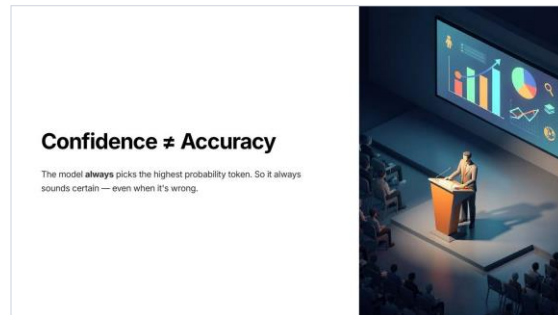
V8 · page 7 · Pattern frequency becomes probability

Chat question 3 of 3

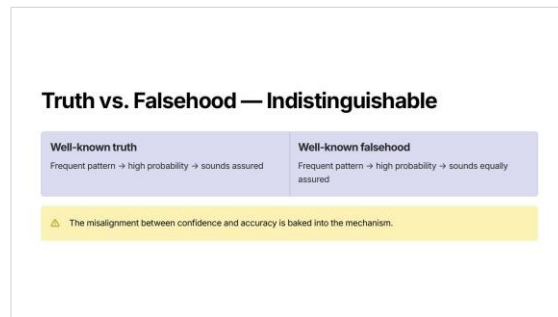
Why does a language model always sound certain, even when it is wrong?

- A. Because it was trained to hide doubt.
- B. Because it always writes the highest-probability token, and a frequent falsehood gets a high probability just as a frequent truth does; nothing in the loop checks facts.
- C. Because it has read more than any person and is therefore rarely wrong.
- D. Because certainty is a setting the vendor turned on.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.



V8 · page 9 · Confidence is not accuracy



V8 · page 11 · Truth and falsehood: indistinguishable

Your turn: the token budget, then the next token

The numbers are the running example. Part one: move the word count and choose a tool's window; the panel converts words to tokens (times 1.33) and says what is left for the answer. Part two: choose a context; the panel shows the fixed probability table and the token the model writes. The checkbox makes a false claim frequent in the training text and shows what it does to the table.

The controls (each with what it stands for)

- Words in the document you paste — 100 to 20000 words, default 10000 words. The length of what you paste, in words. Tokens are words multiplied by 1.33 in this example (about three quarters of a word per token).
- Token window of the tool — options: 8,000 tokens (the deck's 2023 figure); 32,000 tokens; 128,000 tokens; 200,000 tokens; 1,000,000 tokens (current vendor pages, September 2026). The most the tool can hold at once: your prompt, the conversation so far and the answer, together.
- The text so far (the context) — options: “The capital of France is”; “The telephone was invented by”; “Our third-quarter revenue was”. What the model has read when it predicts the next token. The probability table below it is a fixed teaching value chosen to behave like a real run.
- The false claim “Lincoln invented the telephone” appeared 2,000 times in the training text — a tick box, off by default. Only affects the telephone context. Shows what a frequent falsehood does to the probabilities: it becomes the most likely continuation and is written with full confidence.

Try

1. Leave 10,000 words and 8,000 tokens: about 13,300 tokens, over the window; then choose 1,000,000 tokens and read what is left for the answer.
2. Choose “The telephone was invented by”: Bell gets 90% and is written. Tick the checkbox: Lincoln gets 55% and is written instead, with the same confidence.
3. Choose “Our third-quarter revenue was”: the most likely token is a dollar sign; the number that follows is a guess shaped like a number, which is why every figure is checked (Lesson 6).

The point: The budget is arithmetic you can do before pasting; the prediction is a table in which frequency, not truth, decides. The probability tables are fixed teaching values chosen to behave like real runs.

Three things from this block

1 Estimate before you paste: words times 1.33, plus room for the answer, against the window of the tool you use.

2 Every answer is next-token prediction repeated thousands of times: no database, no reasoning engine, no verification step.

3 Prompt engineering is steering probabilities: what you put in the prompt changes what comes out, which is Session 2's subject.

What This Explains



What LLMs do well

Fluent, confident output on common patterns



Where LLMs fail

False claims stated with equal confidence as true ones



Why prompts matter

Context shifts probabilities — your words shape the output

V8 · page 14 · What this explains

One Mechanism. Everything Explained.

Next-token prediction. Probability. Greedy selection. Repeat.

☐ Next: techniques that use this insight to get better answers from AI.



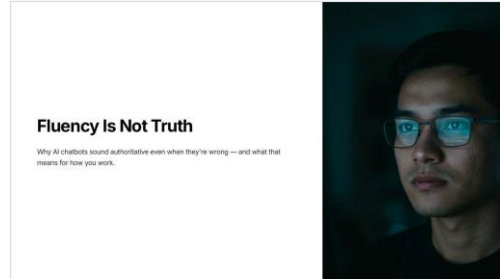
V8 · page 15 · One mechanism, everything explained

Lesson 6

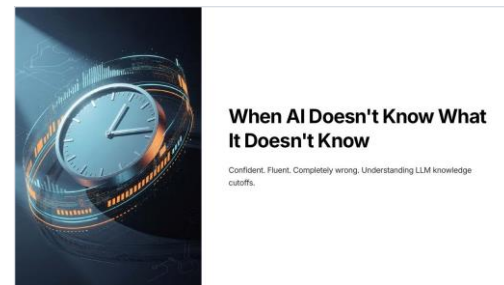
Fluency is not truth: why the same prompt gives different answers, why the model's knowledge stops at a date, and why a hallucination is the mechanism working

6 minutes · decks V9, V10, V11, V12 · shown live: V9

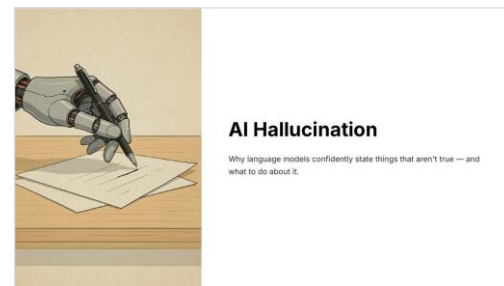
The one idea: Fluency and accuracy are two different skills produced by two different processes; the model was trained for the first, so a beautifully written answer needs the same check as a badly written one.



V9 · page 1 · Fluency is not truth



V11 · page 1 · When AI doesn't know what it doesn't know



V12 · page 1 · AI hallucination

The one idea

Fluency and accuracy are two different skills produced by two different processes; the model was trained for the first, so a beautifully written answer needs the same check as a badly written one.

The founder who never existed

Ask who founded the Institute for Advanced Study and when. The model answers: “Dr. Eleanor Hartwick, in 1937.” Clean grammar, authoritative tone, no hedge. The Institute was founded by Abraham Flexner in 1930; the name is invented and the year is seven years off. Nothing in the text signals the errors, because the text is built to sound right.

the word: hallucination

Four triggers, one dial, one date

Hallucination is most likely on obscure topics, precise numbers, citations and events after the training cutoff, because that is where the training data is thin. Temperature spreads the pick among likely tokens, which is why the same prompt gives different answers. And the model knows nothing after its cutoff date; RAG (retrieval-augmented generation) puts fresh text into the prompt so that it writes from that instead of guessing.

the word: knowledge cutoff

The practical rule of V9 and V12: every specific claim from a language model (a date, a name, a number, a citation, a quote) is unverified until you have checked it against a source outside the model.

Two Separate Skills

Fluency and factual accuracy are not the same thing — and understanding that separation is the foundation of everything else.

Fluency

Grammatically correct, logically coherent, well-structured text that reads smoothly and sounds authoritative.

Factual Accuracy

Correspondence to real-world truth — information verifiable against reliable, external sources.



V9 · page 3 · Two separate skills

A Concrete Example

You ask: “Who founded the Institute for Advanced Study and in what year?”

The AI’s Response

“Dr. Eleanor Hartwick founded the Institute for Advanced Study in 1937.”

Grammatically clear. Totally authoritative. Reads exactly like a reliable answer.

The Reality

The Institute for Advanced Study was founded by **Abraham Flexner** in **1930**. The name “Eleanor Hartwick” is invented. The year is off by seven. Nothing in the text signals these errors.

V9 · page 10 · A concrete example

HALLUCINATION TRIGGERS

When It’s Most Likely to Happen

1

Obscure Topics

Sparse training data → more extrapolation, more guessing.

2

Precise Numbers

Exact figures are rare in training data. Specificity increases risk.

3

Citations & References

Models learn formatting perfectly — not how to verify sources are real.

4

Post-Cutoff Events

No ground truth past training date → invented events that feel authentic.

V12 · page 9 · When it’s most likely to happen

Story: the sentence in the advertisement that cost a hundred billion dollars (February 2023)

1 A launch in a hurry

In early February 2023 Google announced Bard, its answer to ChatGPT, and posted a short advertisement showing the chatbot at work. One example asked what to tell a nine-year-old about the James Webb Space Telescope.

2 The fluent answer

Among the answers: the telescope “took the very first pictures of a planet outside of our own solar system”. Clean, confident, the kind of sentence a science writer would sign.

3 One astronomer reads it

It was false. The first picture of a planet outside the solar system had been taken in 2004 by the Very Large Telescope in Chile, nearly twenty years earlier. Astronomers pointed it out publicly; Reuters reported the error.

4 The market reads it too

On Wednesday 8 February 2023, the day of the launch event, Alphabet’s shares fell about 7.7%, roughly a hundred billion dollars of market value, with the error and the rushed launch widely named as the cause.

5 The lesson the company drew

Google said the episode showed the importance of rigorous testing and opened a testing programme before release. The sentence had every mark of credibility except one: nobody had checked it against a source outside the model.

Sources: CNN Business, “Google shares lose \$100 billion after company’s AI chatbot makes an error during demo” (8 February 2023); Space.com, “James Webb Telescope question costs Google \$100 billion”; both checked 30 September 2026; retold in our own words.

Three things to keep

1 Fluency and accuracy are separate: the model learned how authoritative text sounds, not what is true; confidence is a pattern, not a signal.

2 Different answers every time are sampling and temperature, not malfunction; thin data is where the answers vary most.

3 Knowledge stops at the cutoff; RAG puts fresh text in the prompt; and every specific claim is checked before it is used.

Key Takeaways

- | | |
|---|---|
| 1 Fluency and accuracy are separate
<small>Produced by different mechanisms — a model optimized for one is not automatically good at the other.</small> | 2 Confidence is a learned pattern
<small>Not a signal of correctness. Authoritative tone is reproduced regardless of truth value.</small> |
| 3 Prompting doesn't fix the architecture
<small>Useful at the margin, but the structural possibility of hallucination remains across all model generations.</small> | 4 Verify every factual claim
<small>Dates, names, statistics, citations — independently, every time, before professional use.</small> |

V9 · page 18 · Key takeaways

Key Takeaways

Cutoffs are by design Not bugs — built into how models train	Fluency ≠ Accuracy Confident output can mask stale or invented facts	RAG is the standard Industry solution for real-time reliability
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V11 · page 14 · Key takeaways

Chat question 1 of 3

A junior analyst writes a client memo with a market statistic taken from a chatbot: “The sector grew 14.3% in 2025, according to a 2026 report by the Global Retail Institute.” Using V12’s triggers, how many risk points does the claim carry, and what should happen?

- A. One point (a number); use it, the citation makes it safe.
- B. None: the sentence is precise and cites a report, which is what reliable claims look like.
- C. Three or four points: a precise number, a citation, and possibly an event after the cutoff, on a topic the model may have seen little of; do not use it until two independent sources confirm it.
- D. Ask the chatbot whether the report is real.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

HALLUCINATION TRIGGERS			
When It's Most Likely to Happen			
1	2	3	4
Obscure Topics Sparse training data → more extrapolation, more guessing.	Precise Numbers Exact figures are rare in training data. Specificity increases risk.	Citations & References Models learn formatting perfectly — not how to verify sources are real.	Post-Cutoff Events No ground truth past training date → invented events that feel authentic.

V12 · page 9 · When it's most likely to happen

Red Flags to Watch			
 Overconfident on Niche Topics Certainty about obscure subjects with no hedging.	 Decimal Precision Dates or numbers with suspiciously exact specificity.	 Unfamiliar Citations Sources you've never heard of in a field you know well.	 Recent Event Claims Anything near or past the model's training cutoff.

V12 · page 17 · Red flags to watch

Chat question 2 of 3

A marketing team runs the same product-description prompt three times and gets three different opening sentences. The manager asks whether the tool is broken. What is the right answer?

A. Yes: a working tool gives the same answer to the same question.

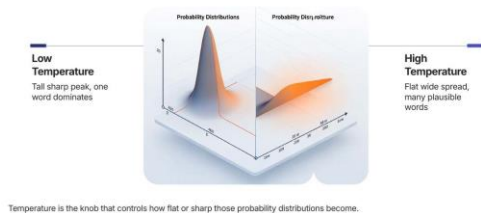
B. No: the tool samples the next word from its probabilities and the temperature is above zero; lower it for consistency or accept the variety and edit.

C. No: the model learned from the first run and improved.

D. Yes: the prompt should be rewritten in capital letters.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

Temperature Reshapes the Odds



V10 · page 4 · Temperature reshapes the odds

Real-World Scenarios



V10 · page 8 · Real-world scenarios

Chat question 3 of 3

A compliance officer asks a chatbot about a rule that changed last month and receives a confident, detailed answer. Which sentence is right?

A. The answer is current, because the model is connected to the internet.

B. The model's knowledge stops at its cutoff date; a change after it is absent, and the model fills the gap with plausible text; the officer checks the regulator's site, or uses a tool that retrieves the current text into the prompt (RAG).

C. The answer is wrong on purpose to avoid liability.

D. Detailed answers are more reliable than vague ones.

Type the letter in the Zoom chat. Countdown: 45 seconds. The answer is discussed in the session.

What Is a Knowledge Cutoff?

The final date included in an LLM's training dataset — a hard stop on the calendar.



GPT-4
April 2024

Claude
April 2024

V11 · page 3 · What is a knowledge cutoff?

RAG

Retrieval-Augmented Generation

Inject current information into the prompt **before** the model generates an answer.



V11 · page 8 · RAG: retrieval-augmented generation

Your turn: turn the temperature, then triage a claim

The numbers are the running example. Part one: move the temperature; the panel reshapes the deck's probability table (the 40%, our 25%, this 15%, another word 20%) and says how many different first words five runs would give. Part two: tick the triggers that apply to a claim you want to use; the panel counts the risk points and states the check required. A live source attached through RAG removes one point.

The controls (each with what it stands for)

- Temperature — 0.1 to 1.5, default 0.7. The dial that reshapes the probabilities before a token is picked: near 0 the top token takes almost all the weight; near 1 the table is used as it is; above 1 the odds flatten and unlikely tokens get real chances.
- Trigger 1: the topic is obscure (little training text about it) — a tick box, off by default. With few examples the model extrapolates instead of repeating; the sparser the data, the more the output is invention.
- Trigger 2: the claim contains a precise number — a tick box, off by default. Exact figures are rare in training text, so the model produces one that feels plausible rather than one it has seen.
- Trigger 3: the claim cites a source (author, report, journal) — a tick box, off by default. The model learned what citations look like, not how to check them; invented references have perfect formatting.
- Trigger 4: the claim concerns something after the model's cutoff date — a tick box, off by default. After the cutoff the model has no data at all and writes what such news usually looks like.
- A live source was retrieved into the prompt (RAG) — a tick box, off by default. Retrieval-augmented generation: current, trusted text is fetched and placed in the prompt before the model answers. It removes one risk point here; it is a mitigation, not a cure, because the model can still misread the retrieved text.

Try

1. Set the temperature to 0.2: the top word gets about 88% and five runs give about one or two different first words. Set it to 1.5: about 35% and about three.
2. Tick triggers 2 and 3 (a precise number with a citation): two points, high risk; the check is two independent sources. Tick RAG: one point, medium.
3. Untick everything: zero points, low risk; read the sentence that still says to check before it goes to a client.

The point: Temperature explains why answers vary; the triggers say where they are most likely to be wrong; neither replaces the check. The probability table and the point scale are fixed teaching values chosen to behave like real runs.

Three things from this block

1 Large language models are fluency engines, not truth engines; treat them as drafting tools whose output is reviewed, every time.

2 Temperature and sampling explain variety; the knowledge cutoff explains silence dressed as an answer; RAG is the standard fix for freshness, and it still needs a check.

3 Build the triage reflex: obscure topic, precise number, citation, recent event; the more of these, the more sources before use.

Use AI as a Drafting Tool

Once you see clearly that confidence and correctness are decoupled, you'll use these tools with much better judgment.

Stop expecting AI to be a truth engine. Start treating it as a drafting tool that requires review.

Understanding the distinction changes how you work — not by avoiding AI, but by building the right habits around it.



V9 · page 19 · Use AI as a drafting tool

Key Takeaways

Hallucination Is Structural

Predictable output of statistical prediction — not a bug.

Confidence ≠ Truth

Phrasing style is not a discriminator of accuracy.

High-Risk Claims

Numbers, citations, obscure topics, recent events.

Always Verify

Treat every specific claim as unverified until checked.

V12 · page 20 · Key takeaways

Same prompt, five outputs

Show, before any theory settles, that the model predicts text rather than looking facts up: the same factual question, asked five times in the same tool, gives answers that differ, and the parts that differ most are the parts that are rarest in the training text. The lab seeds the Module 2 activity (Same Prompt, Five Outputs: Variation Log), which every student completes before Wednesday.

The prompt (copy it exactly): Answer in three lines and nothing else. Line 1: the year the Montréal Metro opened to the public. Line 2: the number of stations it opened with. Line 3: the title, newspaper and date of one newspaper article published about the opening.

The three lines are chosen to sit at three levels of risk. The year is well known and appears in thousands of texts, so it should be stable across the five runs. The station count is a precise number that appears in far fewer texts, so it may move. The article citation asks for something rare and specific, which is exactly where a model fills the gap with a plausible invention (Lesson 6, trigger 3). If your cohort is not in Montréal, replace the Metro with any local landmark that has a well-known year, a less-known count and no famous article.

Your task before Session 2 (Variation Log): Before Session 2: run one factual-recall prompt of your own five times in your tool, in five fresh conversations. Paste each output word for word into the log. Highlight every difference, name at least two places where the outputs contradict each other, and write one paragraph that explains the variation with the mechanism of Lesson 5 (the model gives every next token a probability and picks one) and Lesson 6 (temperature spreads the pick; rare facts have thin data). Submit the log to the portfolio.

Steps in the session

1. Share your screen with the chat tool open in a fresh conversation. Paste the prompt and run it. Read the three lines aloud and type them in the Zoom chat as “Run 1: ...”.
2. Open a new conversation each time (not a follow-up in the same chat, which would let the model see its own earlier answer), paste the same prompt, run it. Repeat until five runs are in the chat.
3. Students: run the same prompt five times in your own tool (new conversation each time) and paste the lines that differ from the professor’s Run 1 into the Zoom chat, with your tool’s name.
4. Read the room: which line changed the least? Which the most? Ask two students by name to say which of the five article citations they would bet is real, then check one citation together in a library catalogue or a newspaper archive search. If none can be found, say so: that is the finding.
5. Close the lab with the two sentences below, and tell students that the Variation Log they submit before Wednesday is this exact exercise with their own prompt and a paragraph explaining the variation with the words of Lesson 5 and Lesson 6.

Case: the summer reading list that did not exist

On Sunday 18 May 2025 the Chicago Sun-Times printed a 64-page special section called “Heat Index: Your Guide to the Best of Summer”. It was not made by the newspaper’s own journalists: it was licensed content from King Features, a syndication company owned by Hearst, and the same section had run in The Philadelphia Inquirer on 15 May. Inside was a summer reading list of fifteen books with authors, titles and one-paragraph descriptions. Ten of the fifteen books do not exist. Real, well-known authors were credited with novels they never wrote, such as a climate novel called “Tidewater Dreams” attributed to Isabel Allende.

Readers noticed within two days. On 20 May the story spread through technology and media outlets, and the freelance writer, Marco Buscaglia, said in public that he had used an AI tool to help draft the list and had not checked the result, calling it an obvious and embarrassing error on his part. King Features said it has a strict policy against creating material with AI, that the writer had used it without telling them, and ended its relationship with him. The Sun-Times said the section had not been reviewed by its newsroom before printing, removed it from the digital edition, replaced it with a letter from its chief executive, and said print subscribers would not be charged for that edition.

On 29 May the Sun-Times published its own review of the section. The reading list was not the only problem: ten of the section’s articles that named sources each contained several errors, including experts who do not work where the section said, a professor the university had no record of, and quotes attributed to real people who never said them. The newspaper announced that it would stop buying special editorial sections from King Features, that it was drafting an AI policy with reader input, and that it would not use AI agents to write its stories. The context matters: the Sun-Times had reduced its editorial staff by about 20% through buyouts shortly before, and the Inquirer said its newsroom is not involved in producing syndicated features it has bought for over forty years.

Poll (type A or B in the chat): whose fault was it?

A: the writer’s fault: he used the tool and did not check.

B: the process’s fault: nobody between the writer and the printing press read the section.

By the numbers

- 15 books recommended, 10 of which do not exist (the Sun-Times review of 29 May 2025).
- 64 pages in the Sun-Times section (18 May); a 56-page version ran in The Philadelphia Inquirer (15 May).
- 2 days between printing and public discovery (18 to 20 May 2025).
- 10 articles with named sources examined in the review; each contained multiple errors (invented experts, misattributed quotes).
- About 20% of the Sun-Times editorial staff had left through buyouts before the incident.
- 0 newsroom reviews of the section before it was printed, at either newspaper.

Why this is not a one-sided story, and the four questions

The easy reading is “the writer used AI and lied”. But the writer did not hide anything from readers: he trusted a fluent list, and nothing in the chain of a syndicate, two newsrooms and a printing schedule read the insert before it reached the public. King Features had a policy against AI; the policy was not a check. The newspapers had bought the section for decades from a supplier they trusted; trust was not a check either. The unresolved tension is whether responsibility sits with the person who pressed the button, with the organisations that removed every reading step to save money, or with both, and in what shares. That is also the question of Session 4, when the review gate is designed rather than discussed.

Discussion questions (asked by name after the poll)

1. The list read perfectly: real authors, plausible titles, a warm paragraph each. Using Lesson 6’s words, why did nothing in the text itself signal that ten books were invented, and what would have signalled it?
2. The writer asked the tool for a list and got fluent text. Which of the four hallucination triggers (obscure topic, precise number, citation, recent event) does a list of new books by known authors hit, and why is that the worst kind of request to trust without checking?
3. King Features had a strict no-AI policy, and the section still went out. What is the difference between a policy and a check, and where in this chain would one fifteen-minute check have stopped the error?
4. The Sun-Times had just lost a fifth of its editors. If you were its chief executive on 21 May, which of these would you do first, and why: stop buying syndicated sections, write an AI policy, or add a review step to everything that carries the paper’s name?

The twelve scenario cards of the Term-Matching Concept Map

Sort each card under the most specific label that fits, with a one-sentence justification; then reconcile with your partner. Labels: Automation (rules, not AI) · AI by rules (AI, not ML) · ML (learns from examples) · LLM (learns from text, writes text).

Card 1

A music app puts together a playlist for you every Monday from what you and people with similar habits played last week.

Card 2 · usually disputed

Your bank blocks a card payment abroad and sends you a text: “Was this you?”

Card 3

You write a complaint in a website’s chat window and a bot answers in three polite paragraphs.

Card 4 · usually disputed

A word processor underlines a sentence and suggests a shorter wording.

Card 5

A thermostat keeps the office at 20 °C: heat on below 20, off above.

Card 6

An e-mail service moves a message you never saw before into the spam folder.

Card 7

Your phone unlocks when it sees your face, even with glasses on.

Card 8

A phone menu says “press 1 for billing, press 2 for technical support” and routes your call.

Card 9 · usually disputed

While you type an e-mail, the editor proposes the rest of your sentence in grey and you press Tab to accept it.

Card 10

A navigation app tells you that you will arrive in 23 minutes, and updates the figure as you drive.

Card 11

A 1980s chess program plays a decent game using thousands of rules written by chess masters.

Card 12

You paste a 20-page report into a chat window and ask for a one-page summary.

Before Wednesday, what is assessed, and when

To do before Session 2 (Wednesday 7 October 2026)

1. Open a free account on ChatGPT, Claude, Gemini or Copilot before Wednesday; keep the same tool for the whole course.
2. Term-Matching Concept Map (Module 1): sort the twelve cards below, justify each in one sentence, reconcile with your partner, submit to the portfolio before Wednesday 7 October.
3. Same Prompt, Five Outputs: Variation Log (Module 2): five runs of one factual prompt in fresh conversations, word for word, differences highlighted, two contradictions named, one explanatory paragraph; before Wednesday 7 October.
4. Quizzes 1 and 2 on the platform before Wednesday 7 October (five questions each, one retake).
5. Watch V13 to V24 (about 70 minutes) and do the Module 3 and 4 readings before Wednesday; the session assumes you have.

Next session. Session 2, Wednesday 7 October 2026, 10:30–12:00 EDT: where bias comes from, and building a prompt one element at a time (V13 to V24)

Component	Weight	What and when	First dates
Attendance and participation	20%	Every session, scored full / half / none	Six sessions, 2 to 21 October
Platform work: quizzes and portfolio	20%	Quizzes 1–2 before 7 Oct; 3–4 before 9 Oct; 5 before 14 Oct. Assignment 1 (prompt log) Mon 12 Oct; A2 (hallucination hunt) Tue 13 Oct; A3 (workflow map) Fri 16 Oct; A4 (two-step chain) Tue 20 Oct; A5 (policy brief) Fri 23 Oct; all 23:59	Concept Map and Variation Log before Wed 7 Oct
Peer learning and feedback	10%	Concept-map pair reconciliation; live peer review of workflow maps (Session 5); capstone feedback forms	Pairs before 7 Oct; forms by Fri 30 Oct
Capstone portfolio	50%	Prompt sequence of three or more steps with an error-recovery step, quality-criteria evaluation, peer-feedback responses, 300-word reflection	Draft and recording Wed 28 Oct; final Wed 4 Nov 2026